



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO N. S. BHARANI FUELS

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

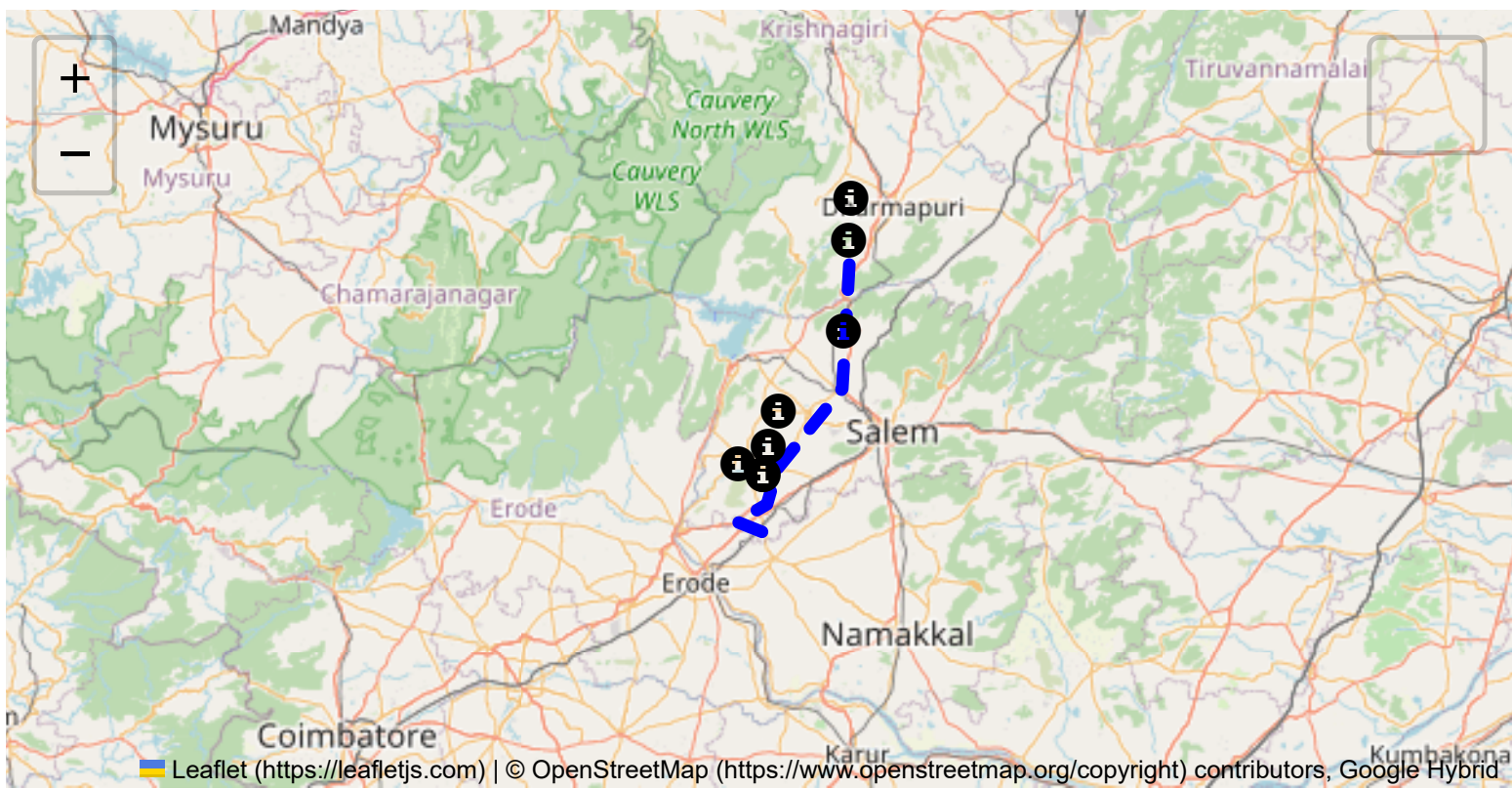
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## **Journey Risk Mitigation**

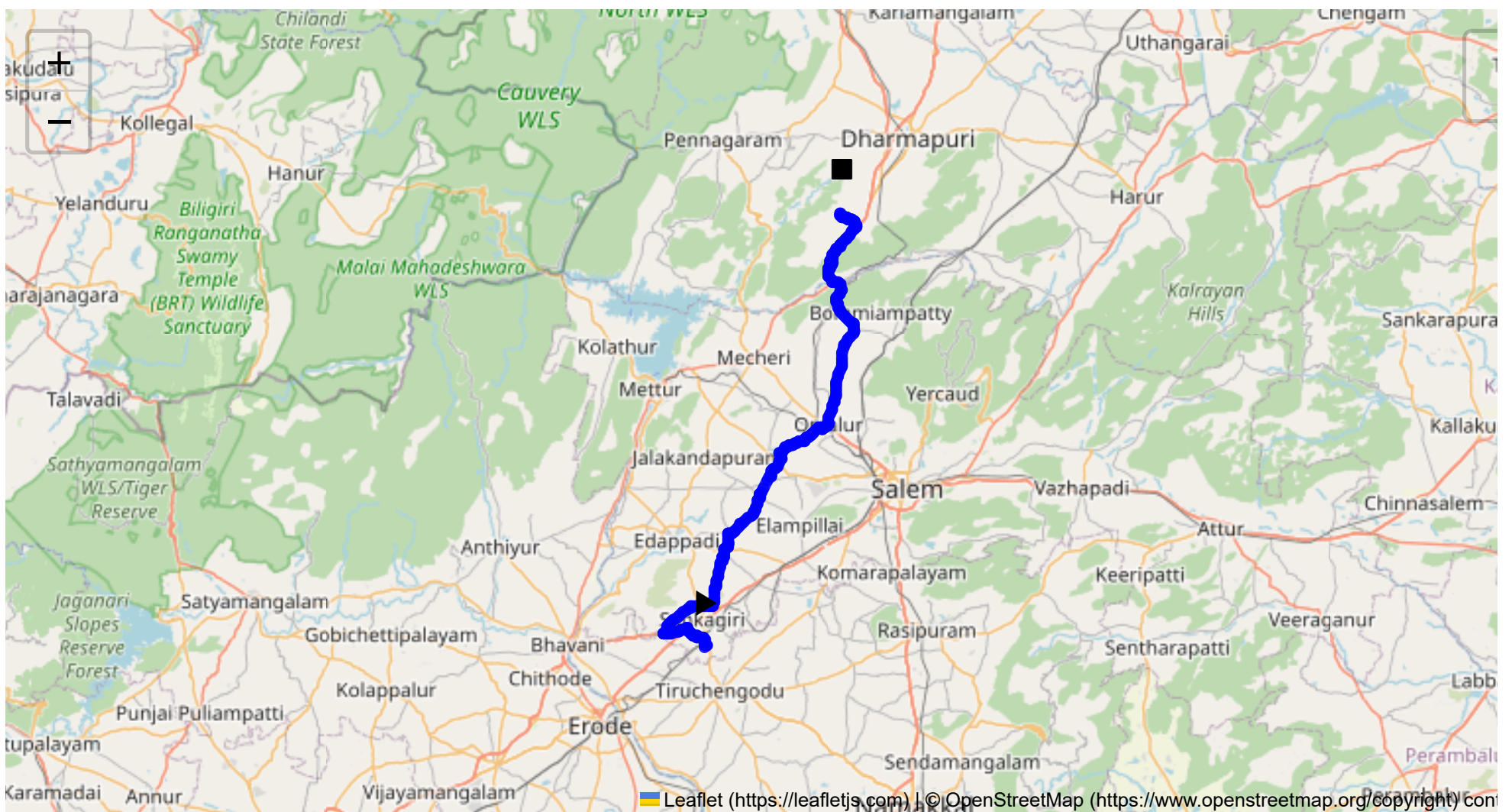
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



**Route Summary:**  
**Total Distance: 92.71 km**  
**Estimated Duration: 2.1 hours**  
**Adjusted Duration (Heavy Vehicle): 2.6 hours**  
**Start: (11.4381, 77.8734)**  
**End: (12.0303464, 78.0630407)**



## Welcome to the Journey Risk Management Study

### 1. Overview of the Route Map

The route begins at Sangagiri and winds through several key points, eventually reaching Samichettipatti. The journey spans approximately 92.71 kilometers. The route traverses towns like Puthur, Padaiveedu, Konganapuram, Omalur, and Thoppur. The major highways involved include the Bhavani Main Road (NH 544H) and Salem - Kochi Highway, along with State Highway 86.



## 2. Typical Weather Conditions and Potential Weather-Related Hazards

The weather in Tamil Nadu can be hot and humid, particularly from March to May. The monsoon season, spanning June to September, can bring heavy rains, potentially causing flooding and reducing visibility. The dry season (October to March) generally offers more stable driving conditions but be wary of sudden thunderstorms.

## 3. Analysis of Traffic Patterns

Traffic congestion is common in towns like Omalur and along major highways. The peak hours are typically from 8:00 AM to 10:00 AM and from 5:00 PM to 7:00 PM, where major congestion can occur. Markets and schools can cause additional local slowdowns. Attention should be paid to intersections and toll booths as they can cause delays.

## 4. Assessment of Road Quality and Infrastructure

While the main highways like the Salem - Kochi Hwy and SH 86 are generally well-maintained, some connecting roads may have potholes or uneven surfaces. Signage is decent on main roads but might be lacking on rural or less-traveled paths.

## 5. Suggestions for Alternative Routes for Emergencies

In the case of major obstructions or hazards, consider using NH 544 for an alternate parallel route eastwards. The local roads connecting nearby towns also offer alternate connectivity but should be approached with caution for heavy vehicles due to their varying quality.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

Transport of hazardous materials must comply with the regulations set by the Government of Tamil Nadu. Standard protocols include compliance with hazardous warning placards, proper documentation, and ensuring the presence of emergency response equipment.

## 7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

There have been incidents involving heavy vehicles where road conditions or human error were factors. Notably, accidents involving chemical spills have heightened awareness around hazardous transport on these routes, emphasizing the need for driver vigilance and proper training.

## 8. Environmental Considerations and Sensitive Areas

The route passes through agricultural land and near water bodies, necessitating extra caution with hazardous materials to prevent contamination. Drivers should exercise caution near fields and adhere to speed limits to minimize dust and noise pollution.

## 9. Analysis of Communication Coverage

Major towns and highways generally have good mobile network coverage. However, rural areas and less traveled routes might experience dead zones. Backup communication devices or route mapping tools are advised.

## 10. Estimated Emergency Response Times for Different Route Segments

- **Urban Areas (Sangagiri, Omalur):** Approximately 15-30 minutes
- **Rural Segments (Between towns):** Approximately 30-60 minutes
- **Remote Areas:** Could exceed 60 minutes

## 11. Overall Summary of Risk Assessment

The route poses a moderate risk for drivers of heavy vehicles carrying hazardous materials. Well-maintained highways offer a generally safe passage, but congestion and varying road conditions in towns and rural areas require caution. Adverse weather conditions during the monsoon season amplify these risks. Continuous driver alertness, adherence to regulations, and preparedness for communication challenges are critical to ensuring a safe journey. Regular reviews of route conditions and emergency protocols should be undertaken to minimize potential hazards.

### Risk Assessment - Turns

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 0  | Turn      | High       | 11.43961, 77.87341 | 15 KM/Hr    |
| 1  | Turn      | High       | 11.44024, 77.87528 | 15 KM/Hr    |
| 2  | Turn      | High       | 11.44898, 77.87410 | 15 KM/Hr    |
| 3  | Turn      | Medium     | 11.45357, 77.85789 | 30 KM/Hr    |
| 4  | Turn      | High       | 11.45352, 77.85698 | 15 KM/Hr    |
| 5  | Turn      | Medium     | 11.45847, 77.85140 | 30 KM/Hr    |
| 6  | Turn      | Medium     | 11.45898, 77.85135 | 30 KM/Hr    |
| 7  | Turn      | High       | 11.46303, 77.84945 | 15 KM/Hr    |
| 8  | Turn      | Medium     | 11.45543, 77.81740 | 30 KM/Hr    |
| 9  | Turn      | High       | 11.45631, 77.81668 | 15 KM/Hr    |
| 10 | Turn      | High       | 11.49383, 77.88442 | 15 KM/Hr    |
| 11 | Turn      | Medium     | 11.56083, 77.89921 | 30 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 12 | Turn       | Medium     | 11.56180, 77.90019 | 30 KM/Hr    |
| 13 | Turn       | Medium     | 11.57048, 77.90224 | 30 KM/Hr    |
| 14 | Turn       | Medium     | 11.58254, 77.90656 | 30 KM/Hr    |
| 15 | Turn       | Medium     | 11.64217, 77.95200 | 30 KM/Hr    |
| 16 | Turn       | Medium     | 11.64712, 77.95118 | 30 KM/Hr    |
| 17 | Turn       | High       | 11.67858, 77.96834 | 15 KM/Hr    |
| 18 | Turn       | Medium     | 11.70062, 77.97961 | 30 KM/Hr    |
| 19 | Turn       | High       | 11.70218, 77.97866 | 15 KM/Hr    |
| 20 | Turn       | Medium     | 11.71917, 78.01123 | 30 KM/Hr    |
| 21 | Turn       | Medium     | 11.72565, 78.01770 | 30 KM/Hr    |
| 22 | Turn       | Medium     | 11.73769, 78.03451 | 30 KM/Hr    |
| 23 | Turn       | Medium     | 11.73785, 78.03551 | 30 KM/Hr    |
| 24 | Turn       | High       | 11.73651, 78.03871 | 15 KM/Hr    |
| 25 | Turn       | Medium     | 11.74265, 78.04710 | 30 KM/Hr    |
| 26 | Turn       | Medium     | 11.78005, 78.05788 | 30 KM/Hr    |
| 27 | Turn       | Medium     | 11.86928, 78.08384 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 11.88009, 78.08164 | 30 KM/Hr    |
| 29 | Turn       | Medium     | 11.91277, 78.05853 | 30 KM/Hr    |
| 30 | Turn       | Medium     | 11.93746, 78.05331 | 30 KM/Hr    |
| 31 | Turn       | Medium     | 11.95803, 78.04964 | 30 KM/Hr    |
| 32 | Turn       | Medium     | 11.95854, 78.04990 | 30 KM/Hr    |
| 33 | Turn       | Medium     | 11.96015, 78.04945 | 30 KM/Hr    |
| 34 | Turn       | Medium     | 11.96370, 78.05142 | 30 KM/Hr    |
| 35 | Turn       | Medium     | 11.96433, 78.05113 | 30 KM/Hr    |
| 36 | Turn       | High       | 11.96542, 78.04938 | 15 KM/Hr    |
| 37 | Turn       | High       | 11.96650, 78.04927 | 15 KM/Hr    |
| 38 | Turn       | Medium     | 11.96804, 78.05080 | 30 KM/Hr    |
| 39 | Turn       | High       | 12.01399, 78.08651 | 15 KM/Hr    |
| 40 | Turn       | Medium     | 12.01914, 78.08227 | 30 KM/Hr    |
| 41 | Turn       | High       | 12.02037, 78.08260 | 15 KM/Hr    |
| 42 | Turn       | Medium     | 12.02084, 78.08193 | 30 KM/Hr    |
| 43 | Turn       | Medium     | 12.02139, 78.08167 | 30 KM/Hr    |
| 44 | Blind Spot | Blind Spot | 12.02306, 78.07557 | 10 KM/Hr    |
| 45 | Turn       | Medium     | 12.02328, 78.07564 | 30 KM/Hr    |

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 46 | Turn      | Medium     | 12.02349, 78.07557 | 30 KM/Hr    |
| 47 | Turn      | High       | 12.02465, 78.07231 | 15 KM/Hr    |
| 48 | Turn      | High       | 12.02566, 78.07211 | 15 KM/Hr    |
| 49 | Turn      | Medium     | 12.02707, 78.06514 | 30 KM/Hr    |
| 50 | Turn      | Medium     | 12.03020, 78.06332 | 30 KM/Hr    |

## Emergency Locations

|    | type     | name                               | coordinates            | speed_limit | risk_level |
|----|----------|------------------------------------|------------------------|-------------|------------|
| 1  | hospital | Sri Shanthi Hospital               | 11.569272, 77.901937   | 30 km/h     | Medium     |
| 2  | hospital | Sugumar Hospital                   | 11.7396031, 78.0426955 | 30 km/h     | Medium     |
| 3  | clinic   | R.R. Nursing Home                  | 11.7394129, 78.0439861 | 30 km/h     | Medium     |
| 4  | hospital | Sri Hari Hospital                  | 11.7396726, 78.0440701 | 30 km/h     | Medium     |
| 5  | hospital | Omalur Government Hospital         | 11.74161, 78.044296    | 30 km/h     | Medium     |
| 6  | hospital | Sri Sugam Hospital                 | 11.7429821, 78.0461039 | 30 km/h     | Medium     |
| 7  | hospital | Saravana Hospital                  | 11.7409303, 78.0458308 | 30 km/h     | Medium     |
| 8  | clinic   | Saratha Clinic                     | 11.7422818, 78.0458175 | 30 km/h     | Medium     |
| 9  | hospital | Arun Hospital                      | 11.7429892, 78.0466881 | 30 km/h     | Medium     |
| 11 | clinic   | Sri Meenakshi Nursing Home, Salem  | 11.7449162, 78.0474637 | 30 km/h     | Medium     |
| 12 | clinic   | Divyam Child Care Clinic           | 11.7447135, 78.0473858 | 30 km/h     | Medium     |
| 13 | clinic   | Momota Clinic                      | 11.744982, 78.047397   | 30 km/h     | Medium     |
| 14 | clinic   | Gomathi Clinic                     | 11.7445293, 78.0470401 | 30 km/h     | Medium     |
| 15 | hospital | Vinayaka Child Hospital            | 11.745812, 78.0480225  | 30 km/h     | Medium     |
| 16 | hospital | Dr. G.R.Thilakam Hospital          | 11.746097, 78.04769    | 30 km/h     | Medium     |
| 17 | hospital | Thulasi Hospital                   | 11.7461516, 78.0469984 | 30 km/h     | Medium     |
| 18 | hospital | Amirthalingam Hospital             | 11.7456708, 78.0470548 | 30 km/h     | Medium     |
| 19 | hospital | Children Hospital                  | 11.7458377, 78.0480205 | 30 km/h     | Medium     |
| 20 | clinic   | Shri Ramana Poly Clinic            | 11.7477122, 78.0479293 | 30 km/h     | Medium     |
| 21 | hospital | Dr. Nagaraj Hospital               | 11.7464035, 78.0472401 | 30 km/h     | Medium     |
| 22 | clinic   | Manimuthu Skin Clinic              | 11.7521189, 78.0484058 | 30 km/h     | Medium     |
| 23 | hospital | St. Joseph's Speciality Hospital   | 11.761611, 78.0522344  | 30 km/h     | Medium     |
| 25 | hospital | Anugraha Multi Speciality Hospital | 11.773549, 78.0546408  | 30 km/h     | Medium     |

|    | type     | name                        | coordinates            | speed_limit | risk_level |
|----|----------|-----------------------------|------------------------|-------------|------------|
| 27 | hospital | L.R. Hospital               | 11.8703354, 78.0844554 | 30 km/h     | Medium     |
| 28 | clinic   | JSA Clinic                  | 11.8700619, 78.0844982 | 30 km/h     | Medium     |
| 29 | hospital | Thoppur,Government Hospital | 11.9398824, 78.0551658 | 30 km/h     | Medium     |

### Crowded Spots

|    | type    | name                                                 | coordinates               | speed_limit | risk_level |
|----|---------|------------------------------------------------------|---------------------------|-------------|------------|
| 0  | school  | KRP Matric. Hr. Sec School                           | 11.4546193,<br>77.8142445 | 30 km/h     | Medium     |
| 10 | school  | Fathima Hr Sec School                                | 11.7410508,<br>78.0479272 | 30 km/h     | Medium     |
| 24 | school  | John Britto Matriculation Higher<br>Secondary School | 11.7676907,<br>78.0513092 | 30 km/h     | Medium     |
| 26 | college | Dhirajlal Gandhi College of Technology               | 11.7968014,<br>78.0562095 | 30 km/h     | Medium     |

### Route Photos of Risky Spots



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44024, 77.87528





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44898, 77.87410



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45357, 77.85789





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45352, 77.85698



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45847, 77.85140



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45898, 77.85135





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.46303, 77.84945



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45543, 77.81740





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45631, 77.81668



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.49383, 77.88442



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.56083, 77.89921





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.56180, 77.90019



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.57048, 77.90224





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.58254, 77.90656**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.64217, 77.95200**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.64712, 77.95118**





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.67858, 77.96834



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.70062, 77.97961





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.70218, 77.97866



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.71917, 78.01123



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.72565, 78.01770





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.73769, 78.03451



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.73785, 78.03551





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.73651, 78.03871



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.74265, 78.04710



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.78005, 78.05788





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.86928, 78.08384



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.88009, 78.08164





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.91277, 78.05853



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.93746, 78.05331



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95803, 78.04964





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.95854, 78.04990



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.96015, 78.04945





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96370, 78.05142



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96433, 78.05113



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96542, 78.04938





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.96650, 78.04927



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.96804, 78.05080





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.01399, 78.08651

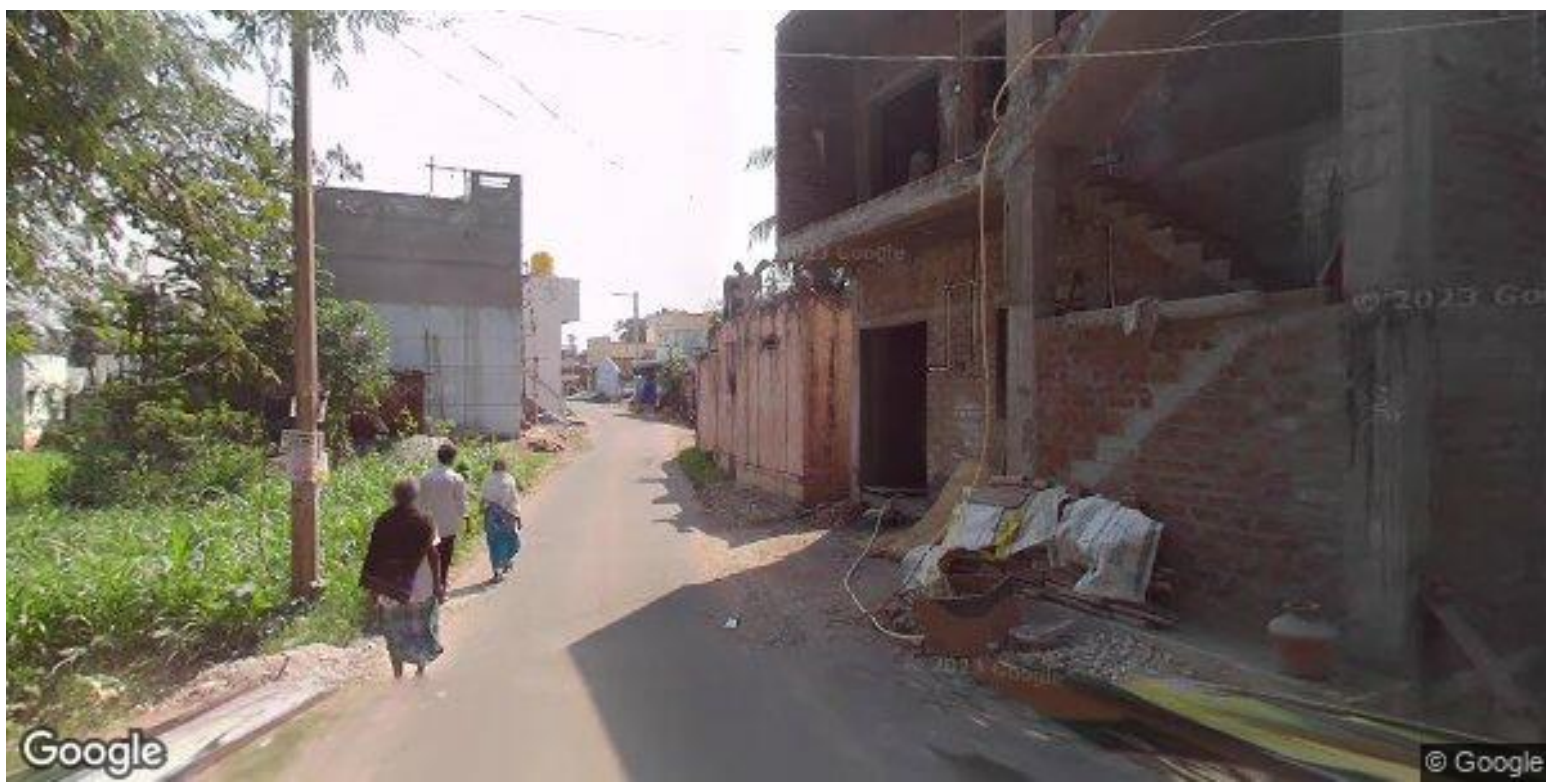


**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.01914, 78.08227



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.02037, 78.08260





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 12.02084, 78.08193



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 12.02139, 78.08167





**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 12.02306, 78.07557



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.02328, 78.07564



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.02349, 78.07557





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 12.02465, 78.07231



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 12.02566, 78.07211





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 12.02707, 78.06514



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 12.03020, 78.06332

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